MAT 238 Calculus 3

3.1 Vector Functions

A **vector valued function** (or just **vector function**) is simply a function whose domain is a set of real numbers and whose range is a set of vectors.

Example A. $\mathbf{r}(t) = \langle 1 + t,\, 2 - t \rangle$



Example B. $\mathbf{r}(t) = \langle 2\cos t,\, \sin t \rangle$



Example C. $\mathbf{r}(t) = (t+1)\mathbf{i} + (t^2+1)\mathbf{j}$

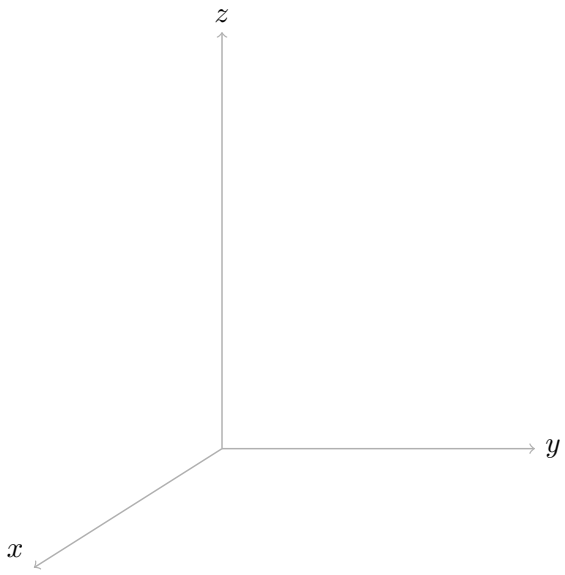


Let $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$. The **curve** C is the set of all points $\big(x(t), y(t), z(t)\big)$ as t varies over the domain of $\mathbf{r}$. The **path** is a particular way of traversing the curve. A **plane curve** is a curve in two dimensions. A **space curve** is a curve in three dimensions.

The **magnitude** of the vector function $\mathbf{r}(t)$ is $\|\mathbf{r}(t)\| = \sqrt{x(t)^2 + y(t)^2 + z(t)^2}$.

Example D. A line in $\mathbb{R}^3$: $\mathbf{r}(t) = \langle 1 + 2t,\, 2 - 3t,\, -6t \rangle$.

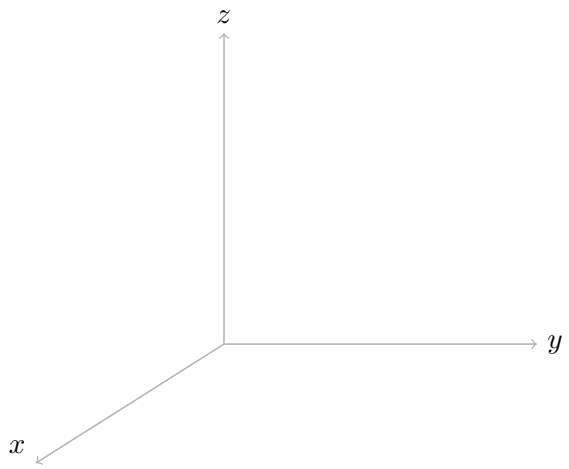
Example E. $\mathbf{r}(t) = \langle 2\cos t,\, 2\sin t,\, t \rangle$



What does it mean to parameterize a curve?

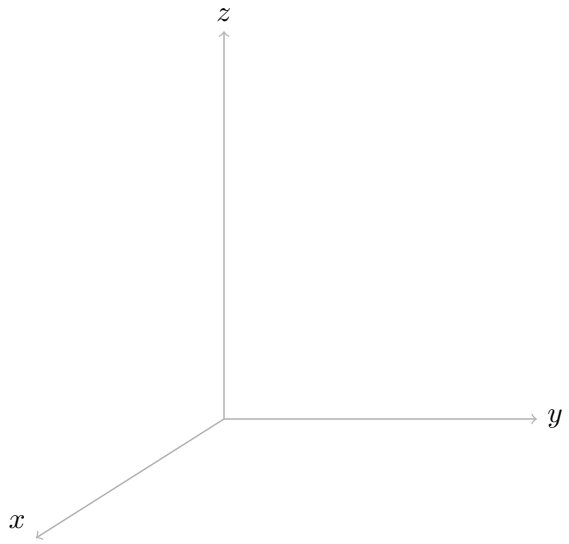
To **parameterize** a curve means to choose a parameter t and express each coordinate as a function of t so that the resulting point moves along the desired curve. The parameter interval determines which part is traced and, together with the coordinate functions, determines the starting point, direction, and number of traversals. We may write the vector equation $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ or the equivalent parametric equations $x = x(t)$, $y = y(t)$, and $z = z(t)$.

Example F. Parameterize the counterclockwise path around a circle of radius 8 that is parallel to the xy-plane and lies 9 units above it. (View counterclockwise from the positive z-axis.)



Note. To find an intersection, both equations must hold at the same time. In practice, this usually means setting two expressions equal to each other or substituting one equation into the other.

Example G. Parameterize the curve of intersection of the cylinder $x^2+y^2 = 25$ and the plane $2x+3y-z = 0$.



Parameterizations to Know

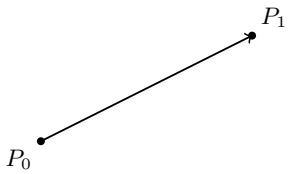
1. Straight line segment from $P_0(x_0, y_0, z_0)$ to $P_1(x_1, y_1, z_1)$

Write $\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$ and $\mathbf{r}_1 = \langle x_1, y_1, z_1 \rangle$.
Vector equation:

$$\mathbf{r}(t) = (1-t)\mathbf{r}_0 + t\mathbf{r}_1, \qquad 0 \le t \le 1.$$

Parametric equations:

$$\begin{cases} x = (1-t)x_0 + tx_1, \\ y = (1-t)y_0 + ty_1, \\ z = (1-t)z_0 + tz_1, \end{cases} \qquad 0 \le t \le 1.$$



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2. Ellipse centered at the origin, width $2a$ and height $2b$

Cartesian equation:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

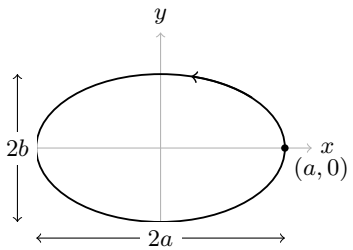
Vector equation:

$$\mathbf{r}(t) = \langle a\cos t,\, b\sin t \rangle, \qquad 0 \le t \le 2\pi.$$

Parametric equations:

$$\begin{cases} x = a\cos t, \\ y = b\sin t, \end{cases} \qquad 0 \le t \le 2\pi.$$

The path begins at $(a, 0)$, travels counterclockwise, and goes around once.



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3. Ellipse centered at (p, q), width $2a$ and height $2b$

Cartesian equation:

$$\frac{(x-p)^2}{a^2} + \frac{(y-q)^2}{b^2} = 1.$$

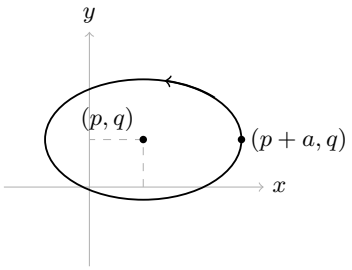
Vector equation:

$$\mathbf{r}(t) = \langle p + a\cos t,\, q + b\sin t \rangle, \qquad 0 \le t \le 2\pi.$$

Parametric equations:

$$\begin{cases} x = p + a\cos t, \\ y = q + b\sin t, \end{cases} \qquad 0 \le t \le 2\pi.$$

The path begins at $(p + a, q)$, travels counterclockwise, and goes around once.



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4. Graph of $y = f(x)$ for $a \le x \le b$

Let the parameter stand for the x-coordinate.
Vector equation:

$$\mathbf{r}(t) = \langle t,\, f(t) \rangle, \qquad a \le t \le b.$$

Parametric equations:

$$\begin{cases} x = t, \\ y = f(t), \end{cases} \qquad a \le t \le b.$$

The graph is traversed from left to right as t increases.

